

CS367 Software Analysis

Intro to Software Analysis

Outline

1. HW1 released - due Friday September 11
2. Unit Testing
3. Exercise to assess prerequisites
 - a. You will need gradescope for this!
 - b. 44VXD6

Unit Testing

Testing

Testing usually involves: **executing** a unit of code, observing its **resulting state**, **asserting** some property on it

Why do we write tests?

1. To find faults
2. To provide documentation
3. To prevent regressions

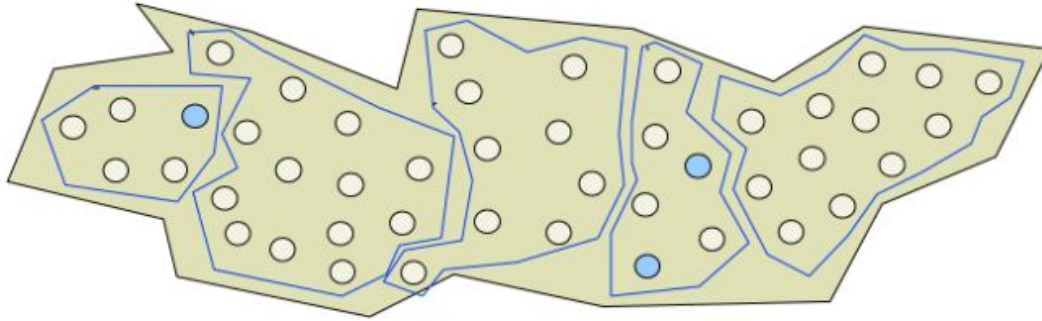
What makes a good test suite?

- Cover all “classes” of input cases
 - Happy cases
 - Invalid cases
 - Edge cases

Equivalence Classes

How can we select inputs?

We can partition the space into “equivalence classes”



What makes a good test suite?

- Cover all “classes” of input cases
 - Happy cases
 - Invalid cases
 - Edge cases
- Diverse
 - Both black-box and white-box criteria
- Executes Quickly
- Covers all portions of the code

What makes a bad test suite?

- Only tests certain classes of inputs
- Redundant
- Slow

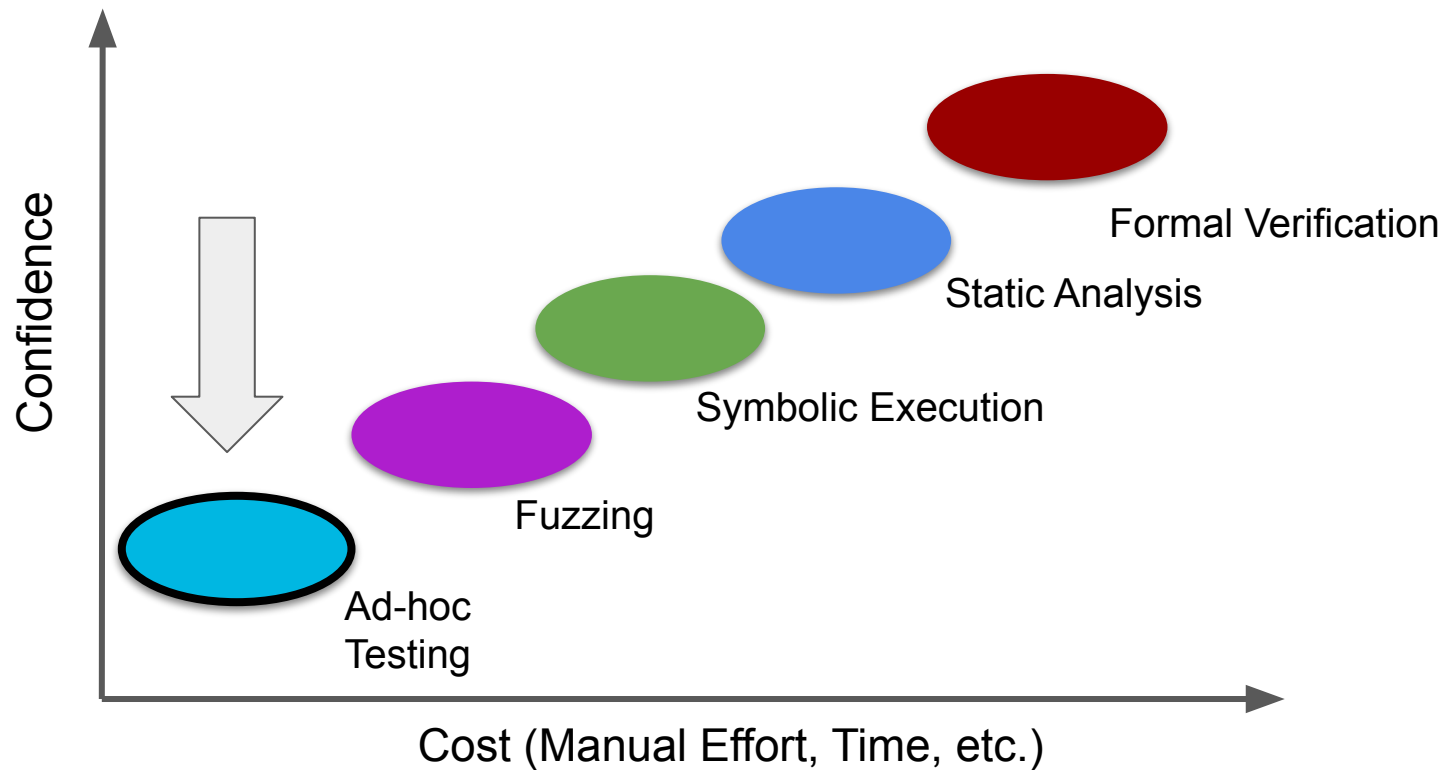
Coverage: a metric for test suite quality

Intuition: untested parts of the code may still contain faults. If your test suite doesn't execute certain sections of the program, potential mistakes in those areas can go undetected

Types of Coverage

1. Line Coverage
2. Statement Coverage
3. Branch Coverage
4. Path Coverage
5. Edge Coverage

Unit Testing has low confidence, but low cost



Exam time!